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(LinkedIn)
github.com/Katy-katy (Portfolio)

Top Skills

Pre-employment Testing
Data Preparation
Reliability Engineering

Languages

English
Russian (Native or Bilingual)

Certifications

Neural Networks and Deep Learning
Apache Drill Essentials
An Introduction to Interactive
Programming in Python
Deep Learning Nanodegree
AI Agents Fundamentals

Honors-Awards

Trainee of the Month Award
CIO Community of Practices
Scholarship

Publications

Utilization of machine learning to
accelerate colloidal synthesis and
discovery

Ekaterina Tcareva

Senior AI Engineer | Production AI Systems | LLMs | NLP | AI Agents
| Generative AI | Retrieval Systems | RAG | Atlanta Metropolitan
Area

Atlanta Metropolitan Area

Summary

Senior Applied AI Engineer with 9+ years of experience building production AI systems, conversational AI platforms, retrieval systems, AI infrastructure, centralized observability systems at LinkedIn and SLAC National Accelerator Laboratory. Experienced in intent classification, prompt engineering, vector search, retrieval pipeline operations, and cross-functional platform support. Strong background in Python, NLP, Azure Cognitive Services, OpenAI APIs, and production AI infrastructure. Experienced building and supporting scalable AI systems used for internal knowledge retrieval, conversational AI, and engineering productivity workflows. Strong in intent classification and LLM-powered workflows using Azure Cognitive Services and OpenAI models. Specialized in:

- LLM-powered applications
- Retrieval-augmented generation (RAG)
- AI agents and conversational AI
- Vector search and retrieval systems
- Production AI reliability and observability
- Intent classification
- Natural language processing (NLP)

Strong background in Python backend development, NLP workflows, search integrations, AI infrastructure, and cross-functional platform operations. Technologies: Python, Azure, Pydantic, NLP, vector search, Scikit-Learn, TensorFlow, Spark, SQL, Pandas, NumPy. MS in Computer Science (Machine Learning), Georgia Tech. US citizen, authorized to work in the US without sponsorship.

Experience

LinkedIn

7 years

Sr. Data Scientist, Applied Science (Sr. AI Applied Engineer)
August 2023 - Present (2 years 11 months)

United States

Designed, built, and operated production AI systems supporting LLM-powered applications, conversational AI, retrieval-augmented generation (RAG),

vector search, enterprise knowledge discovery, and AI agent workflows.

Led initiatives spanning LLM integration, evaluation, observability, platform modernization, reliability engineering, and operational excellence.

- Applied few-shot prompting techniques with OpenAI LLMs to improve intent recognition, response routing, and handling of ambiguous user requests.
- Built and supported production AI applications leveraging large language models, retrieval systems, vector search, and agent-driven workflows.
- Developed evaluation frameworks, test datasets, and quality measurement processes using offline benchmarks, prompt experimentation, production telemetry, user feedback, and error analysis to improve model quality and user experience.
- Implemented production AI operational practices including observability, monitoring, deployment validation, logging standardization, incident management, alerting, and reliability engineering.
- Built and enhanced chatbot and retrieval workflows, including onboarding experiences, FAQ curation, prompt guidance, retrieval optimization, evaluation improvements, and operational support.
- Led migration of curation infrastructure to Python 3.12, validating pipelines and coordinating deployment activities.
- Modernized vector search and retrieval services through schema standardization, Pydantic 2 migration, deployment improvements, and operational maintenance.
- Standardized logging and observability integrations across AI platforms, including migration from OpenTelemetry to Azure Monitor.
- Designed dashboards, monitoring workflows, and alerting systems to improve visibility into service health, performance, and reliability.
- Collaborated with infrastructure, platform, and SRE teams to resolve production incidents and strengthen operational processes.

Data Scientist (NLP, AI Applied Engineer)

January 2020 - August 2023 (3 years 8 months)

United States

Built and scaled a production ticket-classification platform from initial model development through enterprise adoption, ultimately transitioning ownership to a dedicated team after establishing mature operational processes, monitoring, and support workflows.

* Co-created and maintained a production ticket-classification platform built with FastAPI and FastText, enabling automated routing of internal IT and engineering requests

- * Owned the end-to-end machine learning lifecycle, including data cleaning, taxonomy design, training and test dataset preparation, model training, evaluation, deployment, and production monitoring.
- * Performed ongoing misclassification analysis to identify failure patterns, improve training data quality, refine classification taxonomy, and guide model enhancements, increasing classification accuracy from approximately 68% to 94% over three years.
- * Built observability dashboards (Grafana) and monitoring solutions to track model quality, service health, and operational metrics in production environments.
- * Mentored and trained another engineer to assume ownership of the platform, successfully transitioning the mature production system to a dedicated team for ongoing development and support.
- * Developed and supported NLP classification models, Azure Cognitive Services workflows, QnA Maker solutions, and conversational AI systems serving multiple engineering organizations.

Sr. Software Engineer, Machine Learning (NLP)

July 2019 - January 2020 (7 months)

Sunnyvale, CA

- * Worked on NLP and machine learning systems supporting internal conversational AI platforms.
- * Contributed to chatbot infrastructure, classification workflows, and AI platform operations.
- * Supported backend development and ML integration efforts for internal tools.

SLAC National Accelerator Laboratory

Machine Learning/Data Scientist

July 2017 - July 2019 (2 years 1 month)

Menlo Park, CA

- * Co-created Saxkit - Python module for small angle x-ray scattering data analysis
- * Developed clustering workflows for scientific data segmentation and analysis.
- * automated creating of machine learning models for classification of small-angle x-ray scattering spectra (Python, Scikit-Learn, Pandas)
- * Built automated ML pipelines for classification of scientific spectra data (regression models using Linear, Lasso, and Ridge regression techniques for prediction of continuous values for small-angle x-ray scattering spectra)
- * developed API interaction (getting data from a NoSQL database, applying of pre-trained models for predictions)
- * developed user documentation for SASXKIT in HTML format using Sphinx

Jack Baskin School of Engineering, UC Santa Cruz
Implementation of ML, AI, and Data Science projects
September 2015 - June 2017 (1 year 10 months)
Santa Cruz, CA

The projects are listed in the "Projects" section.

NuLeaf Technologies Startup at NASA AMES ASL Labs
member of the core database team
July 2015 - August 2015 (2 months)
Mountain View, CA

Developing a database using MySQL.
<https://www.nuleaftech.com/>

Foothill College
Office Assistant
September 2013 - June 2014 (10 months)
Los Altos, CA

Performed data analysis using Excel.

North West State Technical University
Education Support Specialist, Data Analyst
September 2007 - January 2012 (4 years 5 months)
Saint-Petersburg, Russia

Collected and analyzed students performance data using Excel. Checked students books for corresponding to their education's goal, evaluated sets of tests for corresponding to students' books.

Education

Georgia Institute of Technology
Master of Science - MS, Computer Science · (2018 - December 2022)

University of California, Santa Cruz
Bachelor of Art - BA, Computer Science · (2015 - 2017)

Foothill College
Associated Degree - AS, Computer Science · (2013 - 2015)